

Toy World – E-commerce-Database-Management-System

Chapter 2

2.0 Introduction

The entities required for the Toy World – E-commerce Database Management System are identified based on the business requirements of an online toy shopping platform. The attributes of each entity, along with their Primary Keys (PK), Foreign Keys (FK), Unique Keys (UK), and attribute types, are determined to ensure proper data organization. The relationships among the entities are identified to represent customer interactions, product management, ordering, and payment processes within the system. The cardinality and participation constraints are also analysed to define how entities are connected. This entity analysis serves as the foundation for designing the Entity Relationship (ER) Diagram, relational schema, and database implementation while maintaining data integrity, consistency, and minimizing redundancy.

2.1 Entity and their Attributes and Keys

i. Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
Name	Not Null
Email	Unique Key
Phone	Unique Key
Address	Not Null

ii. Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

iii. Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Not Null

Price	Not Null
Stock	-
Brand	-
Description	-

iv. Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-
Total_Amount	-

v. Entity: Order_Item

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-

vi. Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_Items	-
Total amount	-

vii. Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

viii. Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-
Review_Date	-

ix. Entity: Shipment

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	-
Shipment_Date	-
Delivery_Date	-
Shipment_Status	-

2.2 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Category	Contains	Product	One-to-Many	Partial / Total
Cart	Contains	Product	Many-to-Many	Partial / Partial
Order	Includes	Order_Item	One-to-Many	Total / Total
Product	Appears In	Order_Item	One-to-Many	Partial / Total
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total